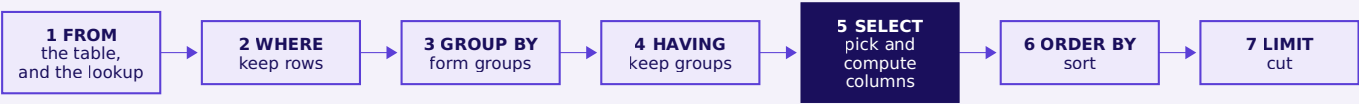


Can the warehouse itself give Anand the Monday numbers, every segment, every week?

Anand Iyer, Kalpa Retail's finance controller, wants last week's revenue tree every Monday for every segment, computed from the warehouse itself, and his analyst audits every line. The book holds 1,000 orders over Q1 (April to June 2026) and Q2 (July to September 2026), each at its amount, whatever its status.

1 In what order does the database run a query's clauses?



A query is written SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY, LIMIT and runs as the arrows show. WHERE tests rows before groups exist, so a test on a count goes in HAVING; SELECT comes fifth, so it cannot print a column the groups do not pin down; LIMIT cuts after ORDER BY. The sheet is Week 1's tree: customers who bought, times orders per customer, times revenue per order, every leaf from orders and the segment from customers. Each panel below is one chapter's check.

2 Which count is a customer, and where should the Monday number be computed?

Query	On the book	It counts
count(*) over orders	1,000	Order rows
count(DISTINCT customer_id) over orders	301	Customers who bought
count(*) over customers	340	Members Kalpa holds

Per quarter, count(*) AS customers printed 538 and 462 at 1.00 order each; counted once, 244 and 227 bought. A .sql file sends back 2 rows where an export copies 1,340.

Crux. A count says what it counts: order rows are count(*), customers are count(DISTINCT customer_id).

3 Does a total that matches last week's mean the story matches?

Q2 over Q1	Customers	Frequency	Order value	Revenue
Last week's extract	1.000	0.860	1.144	0.984
The warehouse	0.930	0.923	1.146	0.984

Both fell 1.6 percent, yet 69 of 69 extract customers bought in both quarters against 170 of 301 in the book, where 7.0 percent fewer bought in Q2 and each ordered 7.7 percent less often.

Crux. A matching total is one leaf matching, so compare every leaf as a change.

4 Why did 140 orders over 76 customers print 1?

```
count(*) / count(DISTINCT customer_id) -- 2, then 1
round(count(*)::numeric
/ count(DISTINCT customer_id), 2) -- 2.36, 1.84
HAVING count(DISTINCT customer_id) < 30 -- Student
```

Two whole numbers divide as whole numbers, truncating towards zero. Multiply back: 1 x 76 is 76, where the orders are 140. Retail-Plus revenue fell 29.4 percent and its frequency 22.0 percent, where the whole numbers said it halved.

Crux. Divide in numeric, round on purpose, and keep the counts beside the ratio.

5 Who is inside the average of member spend?

Retail-Plus, per member	Q1	Q2	Change
avg, a CASE with no ELSE	Rs 6,437	Rs 5,439	-15.5%
coalesce(..., 0) written in	Rs 5,474	Rs 3,863	-29.4%

The first averages 91 members in Q1 and 76 in Q2, since avg skips NULL; the second averages the same 107 in both. Named steps, WITH book AS (...), q1 AS (...), read top down and rerun anywhere.

Crux. An average names who is inside it, so write the zero on purpose.

6 Which numbers add across quarters, and which are counted again?

Retail-Plus in the half-year	Customers
Q1 plus Q2, added	167
Members on the book	120
Counted from the orders	107
Bought in both quarters	60

91 + 76 - 60 = 107. Segments add to the book, 36 + 102 + 91 + 15 = 244, because a customer sits in one segment; quarters do not, because a customer can buy in both.

Crux. Orders and rupees add across quarters; customers are counted again from the orders.

7 Will next Monday's run draw the same five orders?

The fingerprint holds seven numbers: orders has 1,000 rows, 301 customers, a latest date of 28 September 2026 and Rs 19,84,00,000 in all, and customers has 340 rows, 340 customers and a latest joining date of 25 December 2025. On Monday LIMIT 5 with no ORDER BY drew five orders worth Rs 3,900, and after a reload that changed no value it drew Rs 4,590. ORDER BY order_id LIMIT 5 draws the same five, Rs 3,900, both times.

Crux. Order every list on a column no two rows share, and print the book's fingerprint beside the numbers.

THE WORDS ON THIS SHEET

The book The warehouse's two quarters of orders, the one copy everybody reads
Customers who bought Customers with an order in the window, each counted once

HAVING Keeps or drops groups after GROUP BY has formed them
Integer division Two whole numbers divided to a whole number, cut towards zero

CTE A named step written with WITH, which later steps can read
Fingerprint Seven numbers about the book, printed beside each run's numbers

Booked revenue Every order at its amount, whatever its status
Run order FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY, then LIMIT